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A Model of Perceived Risk and Intended Risk-handling Activity

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On the basis of previous reviews of the perceived-risk concept, a model of risk perception and its effects on consumers' risk-handling behavior is formulated. Hypotheses derived from this model are empirically tested and indicate that, for the setting studied (women purchasing a dress), the intended use of risk-handling activity increases with higher levels of perceived risk. This relationship is more pronounced after the level of risk exceeds the individual's acceptable level of risk. Also of importance in determining the use of a risk-handling activity are the perceived benefit of the type of risk-handling activity and the consumer's inability to absorb a monetary loss.

The concept of "risk" became popular in the field of economics in the 1920s (see, e.g., Knight 1921). Since then, it has been successfully used in theories of decision making in economics, finance, and the decision sciences. Bauer (1960) introduced the concept of "perceived risk" to the marketing literature, and then the psychologists Kogan and Wallach (1964) wrote their major work on risk taking. Early work in marketing included a book on risk taking and information handling in consumer behavior (Cox 1967), followed by several conceptual models of consumer risk perception and handling (see, e.g., Markin 1974; Stem, Lamb, and McLachlan 1977; Taylor 1974). More recently, perceived risk has been used as an explanatory variable in empirical research on consumer behavior (see, e.g., Rao and Farley 1987; Srinivasan and Ratchford 1991).

Reviews of perceived risk illustrate a variety of conceptualizations of the perceived risk construct (Dowling 1986; Gemunden 1985; Ingene and Hughes 1985; Ross 1975). The concept of perceived risk most often used by consumer researchers defines risk in terms of the consumer's perceptions of the *uncertainty* and adverse *consequences* of buying a product (or service). In this way, consumer researchers implicitly assume that both the probability and the outcome of each purchase event are uncertain.

This article is also concerned with perceived risk. However, our interest is not only on the determinants of perceived risk but also on how consumers' overall perceived risk (OPR) affects their information-search behavior. This motivates us to develop a conceptual model that contains three general parts. The first delineates a number of factors that can influence a consumer's perceived risk. The second portion of our model partitions this risk into two components, one that captures the person's perception of the riskiness of buying "an average product" in the product class and the second that reflects the perceived risk of the specific alternative being considered. The third portion of our model specifies how these two components of perceived risk differentially affect consumers' information-search behaviors. We then test parts of this model by using a novel two-stage research design. In the first stage, we develop new measures of risk perception, which are used in the second stage to explain differences in the intended search behavior of respondents purchasing a number of items in one product class.

Our findings, in contrast to those of Gemunden (1985), who found that perceived risk is often not a major influence on information search, clearly support the hypothesis that perceived risk influences intended search behavior. As such, these results extend the cross-category findings of Beatty and Smith (1987, namely, there is more search activity in high-risk categories) to an in-category setting. We also raise new issues that challenge some of the present conceptualizations of risk perception used by many consumer researchers.

A DESCRIPTIVE MODEL OF RISK PERCEPTION AND REDUCTION

Our model focuses on situations where consumers contemplate the purchase of a product in a known product class. We assume this purchase is involving and

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that the circumstances surrounding the potential purchase are such that when a person contemplates the purchase, this individual has feelings of "uncertainty," "discomfort," and/or "anxiety." In this way we are in concert with others who use terms such as "concern," "psychological (dis)comfort" (Zaltman and Wallendorf 1983), and "feeling uncertain" (Engel, Blackwell, and Miniard 1986) when talking about perceived risk. We postulate that this type of feeling is the result of the consumer's perception of risk.

In developing our model of the determinants of perceived risk, we acknowledge the work of researchers such as Bettman (1970, 1971), Cox (1967), and Peter and Tarpey (1975) and assume consumers think of perceived risk, at least in part, in terms of the magnitude of consequences and the probabilities that these consequences may occur if the product is acquired. We also postulate that these two aspects of a purchase situation are mapped into the construct of perceived risk. Because product evaluations normally involve both positive and negative outcomes (Cox 1967), we allow for the possibility that both positive and negative consequences will influence perceived risk.

We acknowledge that the consumer's involvement with the purchase decision influences the person's perception of risk. This is in concert with the observation that purchase situations perceived to be of moderate to high involvement are ideal for the study of consumer risk perception (Ingene and Hughes 1985; Oglethorpe and Monroe 1987). Three types of involvement are relevant in our study: ego involvement (Sherif and Cantril 1947), purchase involvement, and product involvement (Bloch, Sherrell, and Ridgway 1986). Ego involvement focuses on the relationship of a product to a person's self-concept, while product and purchase involvement focus on a product category and a specific purchase occasion.

Our model of risk perception is initialized when the consumer first decides to evaluate a product in a known product category for a specific usage situation. Our basic premise is that consumers evaluate the attributes of the specific product being considered, the relevant factors associated with the usage situation relative to their purchase goals and what they know about products of this type (Dowling 1986). Together with the involvement of a person, these factors are mapped into a set of uncertain consequences (hereafter referred to as overall perceived risk) that cause the consumer to feel "uncomfortable." In the third stage of our model we will discuss how the consumer copes with this feeling.

The above-mentioned determinants of perceived risk are shown in the top portion of Figure 1. Note that we leave as unspecified how each consumer psychologically combines these factors into an overall level of perceived risk. (With this said, we subsequently describe how we use these factors to manipulate a person's overall perceived risk.)

The second portion of our model is concerned with the two components of the overall level of perceived risk. Somewhat analogous to Bettman's (1973) "inherent risk" and "handled risk," we also partition overall perceived risk into two components. The first reflects the person's perception of risk inherent in purchasing any particular product in a specific product category. (For example, person A, on the basis of his/her knowledge and past experiences, may perceive the purchase of most refrigerators as inherently risky, while person B may perceive such a purchase to be almost riskless.) We refer to this predisposition as a person's product-category risk (PCR) and assume a person with a higher product-category risk will perceive a given situation to be more risky than a person with a lower product-category risk, all else being equal. The second component of the overall level of perceived risk is associated with the particular product being considered in the product class. Thus, some products in a product class may be perceived as comparably safe while others may be perceived as risky. We refer to this second component of overall perceived risk as product-specific risk (SR).

The specific antecedents for both components of overall perceived risk vary from situation to situation. However, factors that immediately come to mind are (*a*) the levels of the attributes of the specific product considered (e.g., the product's price, quality rating, etc.), (*b*) the likelihood of "failure" that leads to negative consequences, (*c*) the individual's purchase goals (e.g., use for self or as a gift), and (*d*) other conditions associated with the specific purchase situation, such as whether the product is being sold via a catalog or in a retail store.

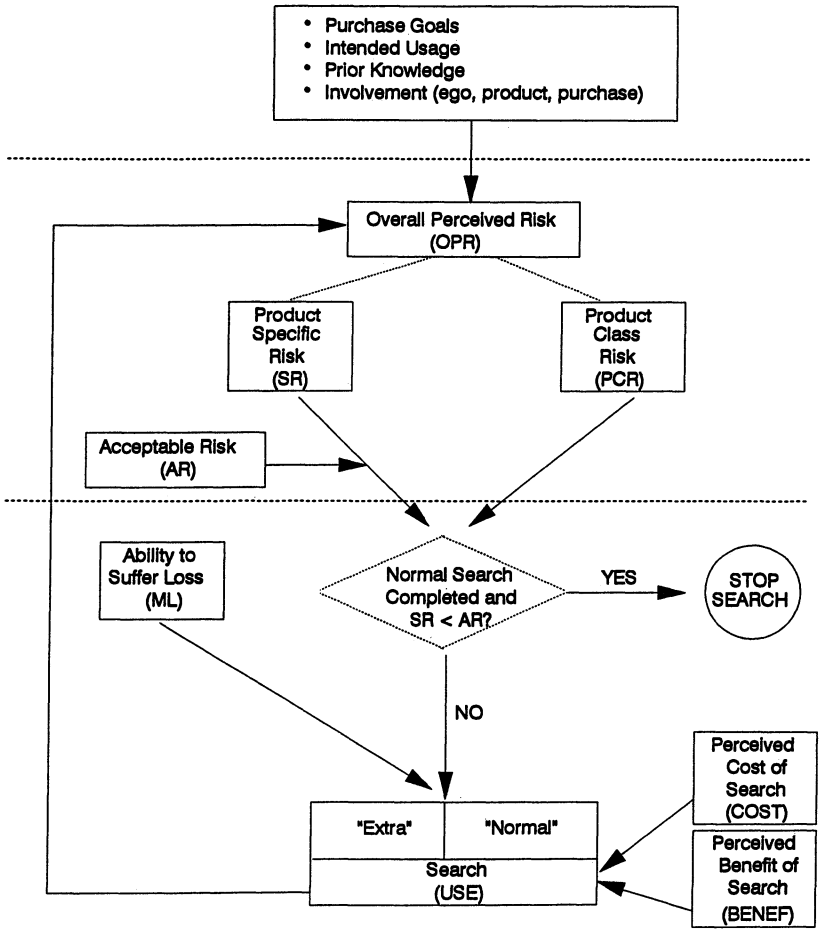
A third major component in this second portion of our model is the concept of an acceptable risk (AR). A growing body of research in psychology, marketing, and management seeks to explain the relationship between risk perception and the acceptance of risk (see, e.g., Kahneman, Slovic, and Tversky 1982; MacCrimmon and Wehrung 1986; Oglethorpe and Monroe 1987). Acceptable risk can be of two types, namely the point above which the product category is perceived as too risky to indulge in (e.g., skydiving, motor racing) and the point above which a specific product has an unacceptable level of perceived risk to purchase. We focus only on the second type of acceptable risk because our model is tested on products in a single product category. We acknowledge that this acceptance of risk may be inseparable from risk perception (Covello, Menkes, and Nehnevajsa 1982). In fact, we subsequently hypothesize that individuals with high levels of product-category risk will have lower values of acceptable risk.

The third portion of our model is concerned with how the two components of perceived risk interact with the person's level of acceptable risk to determine the consumer's information-search strategy. We take as our starting point the concept of problem-focused coping proposed by Folkman and Lazarus (1988). They pro-

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FIGURE 1

PROCESS MODEL FOR PERCEIVED RISK AND INFORMATION SEARCH



pose that when people are in a distressed state (e.g., do not like their jobs) they try to engage in problem-solving activities. Similarly, our model is predicted on the belief that consumers engage in risk-reducing (i.e., information-search) activities in order to reduce their perceived risk level (and, therefore, their feelings of being uncomfortable). We postulate two different generic strategies for accomplishing this risk reduction. First, people routinely conduct information search (or shopping activity) in order to acquire general product-class information. We refer to this type of information search as "normal" risk-reduction activities. The second strategy is to reduce the specific perceived risk of the particular product being considered to an acceptable level. This might be achieved by collecting new information about the brand being considered or other available brands or by modifying the selection of products in the choice set. For example, probability of loss could be reduced by obtaining information from a friend who had used the product, limiting the set of alternatives to well-known brands, or trying the product prior to purchase. Alternatively, the magnitude of the adverse consequences could be reduced by altering purchase goals, and/or by insuring against any adverse consequences (e.g., looking for a product with a warranty or a higher price, thereby "insuring" high quality).

The relationship between our three perceived-risk constructs—product-category risk, specific risk, and acceptable risk—and the person's risk-reduction activities is shown in the lower half of Figure 1. The psychological construct of acceptable risk moderates the relationship between product-specific perceived risk and the use of extra risk-reduction activities. More precisely, when the level of specific risk is less than the acceptable risk level, only normal risk-reduction activities will be pursued. In such a situation, the person "accepts" the specific risk and does not engage in any additional or "extra" risk-reduction activity over and above that which the individual would normally engage in when shopping for a product in this category. However, if the consumer perceives product-specific risk to be unac-

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ceptable, s/he will undertake extra risk-reduction activities in order to reduce this risk.

Figure 1 also indicates that the specific information acquired can alter the individual's perceived risk level and, thus, his/her feeling of discomfort. It is our belief that this person will discontinue search only after (*a*) completing his/her normal search and (*b*) acquiring enough information to reduce the product-specific risk to at least the acceptable risk level. In this way our search model is dynamic and updating.

Finally, our model addresses how specific search activities (e.g., using friends' opinions, etc.) are chosen. We do this by relying in part on concepts derived from economically based search models (see, e.g., Stigler 1961) and the information-processing paradigm (see, e.g., Bettman 1979), both of which suggest specific search activities are selected in part because of their perceived costs and benefits. For example, a person may talk to a friend if the individual perceives the cost of this conversation is less than the benefit (in terms of reducing the person's perceived risk). In addition, we believe person-specific factors other than the individual's product-class risk influence search activity. For example, the person's ability to sustain a loss should alter the amount of effort taken. We control for the above-mentioned factors in our empirical analysis.

To summarize, Figure 1 suggests that, for a particular product, the total amount of perceived risk is composed of the risk associated with buying any particular product in the category and the risk specific to the item being considered. Search activity is entered into with the intent of lowering the person's overall perceived risk level. The identity of the search activities undertaken (and thus the amount of search) is a function of the person's acceptable risk level, the levels of the two components of perceived risk, the costs and benefits of the specific available risk-reduction activities, and the ability of the person to suffer a loss. Specifically, when product-specific risk is less than the person's acceptable risk level (i.e., SR < AR), the person's intention of engaging in a specific type of search behavior is not influenced by product-specific risk. However, the person's intention to engage in a specific type of search behavior increases with (*a*) the perceived benefits (or usefulness) of the search activity, (*b*) the person's product category risk, and (*c*) the person's inability to absorb a loss. Also, the extent of intended search behavior decreases with the person's perceived costs of using the risk-reduction activity. When SR > AR, the probability of engaging in any particular type of risk-reduction behavior increases with the degree to which product-specific risk exceeds the acceptable risk level. Finally, we hypothesize that search will continue as long as SR > AR.

The model outlined in this section is compatible with the conflict theory model of decision making proposed by Janis and Mann (1977), the information-processing paradigm of consumer choice adopted by Bettman (1979), and the economically based search models of Stigler (1961). It advocates that the consumer's intensity of search activities (both normal search activity and extra risk-reduction activity) depends on three individual-level general constructs—product-category risk, acceptable risk, and inability to sustain a loss—a number of situation-specific factors (e.g., attributes of the product, the purchase situation, the person's involvement, and the consumer's goals) as captured by the specific risk-variable SR, and the perceived costs and benefits of each risk-reduction activity.

The model is dynamic in nature in that the level of perceived risk is viewed to change over time as the consumer acquires new information. In the empirical study reported next, we do not track the changes in time for a particular intended purchase. Instead, we conduct a within-subject experiment where we vary the level of situation-specific factors (and, thus, the perceived risk) and then determine the extent to which the person's *intended* search behavior varies according to the major constructs of our model. We note that a more complete testing of the model would determine the effects over time of a person's information-search activities on his/her subsequent perception of risk and purchase behavior. That is, does the person actually have a lower risk level after using various risk-reduction activities, and does a person only consider consummating the purchase after getting the level of specific risk below acceptable risk? Although we do not test these components of the model, we discuss them again in a later section.

The specific hypotheses tested are as follows:

H1: Consumers' total amount of intended risk-reduction activity for a product is increased by
- (*a*) their purchase category risk PCR level
- (*b*) the monetary loss (ML) associated with the intended purchase occasion
- (*c*) the perceived benefits (BENEF) of using various available information-search and risk-reduction activities

and decreased by
- (*d*) the perceived costs (COST) of using various available information-search and risk-reduction activities.

H2: When the level of product-specific perceived risk is acceptable, then it has no significant effect on the total amount of intended risk-reduction activity.

H3: If the amount of product-specific perceived risk is greater than the level of acceptable risk, then the individual's intentions to undertake more risk-reduction activities monotonically increases with SR.

H4: The individual-level constructs of acceptable risk, product-category risk, and ability to suffer a monetary loss are interrelated. Specifically, individuals with less ability to suffer a

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monetary loss and a higher level of product-category risk will have a lower acceptable risk level.

Thus far, the development of the model is entirely in terms of theoretical constructs. To test the model requires operationalizing the various hypothetical constructs in terms of observed variables and specifying the functional form of these relationships, which results in what we term an empirical model. In the next section we present this empirical model. However, before doing so, we present a brief overview of the research design used to collect the data and develop measures for the variables of interest in the model.

THE EMPIRICAL MODEL

Overview of Research Design

The research setting examines the risk perception and intended risk-reduction strategies for the purchase of new dresses. The research design is partitioned into two stages. In the first stage we develop, for each subject, estimates of all the constructs hypothesized to affect intended risk-reduction behavior, namely, product-category risk, product-specific perceived risk, monetary loss, acceptable risk, and the cost and benefit of such activity. In the second stage, we determine the subject's intended behavior for a set of different perceived risk situations. We did not directly measure any of the constructs hypothesized to affect this intended risk-reduction behavior in this second stage. Instead, we inferred measures of these constructs from the estimates developed in the first stage.

More specifically, in the first stage subjects are asked to rate the overall perceived riskiness of a number of different hypothetical dress-purchase scenarios. Each scenario is characterized in terms of the same six attributes of the dress (e.g., price, quality, fit with subject's self-image, etc.) and a fixed purchase goal and purchase situation. With the use of a conjoint methodology, part-worths (i.e., risk utilities) are estimated for each potential consequence for each product attribute for each individual. This allows estimation of overall perceived risk for any type of product described by these product attributes. Also, it allows us to estimate both product-specific perceived risk and product-category risk. In addition, each individual's acceptable risk level is estimated by asking subjects to specify if they would collect more information prior to purchasing the dress described in each hypothetical purchase situation and then relating this to the estimated product-specific perceived risk level for that situation. Finally, subjects respond to a set of questions aimed at measuring their ability to suffer a monetary loss and the perceived costs and benefits associated with the different risk-handling strategies available for use in stage 2.

Stage 2 presents a different task to the subjects. They are given three sets of magazine photos of women's dresses and told to select two dresses from each group. For each of the six dresses chosen, respondents answer a set of questions about their likely risk-reduction shopping behavior (USE), with the assumption that they decided to pursue purchasing that specific dress. Although the intent of this task is to simulate a "typical" shopping process, the measure of information search and other shopping activities obtained is what respondents think would be their behavior and is not their actual behavior.

We focus on intended rather than actual behavior for two reasons. First, our model specifies that actual search behavior is affected by the individual's current perceived risk level and that this level can change with each new piece of acquired information. Thus, the only search measure compatible to initial perceived risk is *initial* (i.e., intended) risk-reduction behavior. The second reason is more practical. To relate actual behavior to perceived risk requires a longitudinal design and repeated measures of perceived risk. Such a design is more complicated than the cross-sectional approach used here. Moreover, it is not needed to test the hypotheses outlined earlier.

Hypotheses 1, 2, and 3 are tested by relating the respondent's intended risk-reduction activities, as measured in stage 2, to their perceived risk levels (SR, PCR, AR), inability to absorb a monetary loss, and perceptions of the costs and benefits associated with using a particular search strategy, as measured in stage 1. In this way, we decouple the measurement of the dependent variables from the predictor measures, thereby reducing possible demand effects. We believe such a procedure provides a strong test of our model. The next subsection presents a set of equations that outline the empirical model.

Operationalizing the Risk-handling Model

Hypotheses 1, 2, and 3 specify the direction of the relationships between the independent measures and the probability of engaging in a particular risk-reduction and shopping activity. However, to test the significance of these relationships, we need to postulate a specific system of equations (and at times their functional forms). We assume the following set of relationships.

$$\text{PCR} = f1 \text{ (individual level variables, attributes of the product class)}, \quad (1)$$

$$\text{SR} = f2 \text{ (purchase goals, purchase situation, specific product attributes)}, \quad (2)$$

and

$$\text{OPR} = \text{PCR} + \text{SR}. \quad (3)$$

Since the analysis is done for each subject and the product class is fixed, PCR is assumed to remain constant for an individual across the dress-purchase situ-

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ations evaluated in the first stage of our data collection. However, SR will vary for each individual because the specific product attributes are different for each scenario. We expect the respondent's OPR ratings to decompose into a fixed PCR component and a variable SR component, where the variable component is a function of the varying product-specific attributes.

Hypothesis 4 states that AR is negatively related to PCR and ML. We test this hypothesis by regressing AR against the other two individual-specific constructs.

We test Hypotheses 1, 2, and 3 simultaneously as follows. First, we define the intention of person i to use risk activity k for dress j to be USE_{ijk}. Then, these three hypotheses imply

$$\text{USE}_{ijk} = (b_0 + b_1\text{ML}_i + b_2\text{PCR}_i) - b_3\text{COST}_{ik} + b_4\text{BENEF}_{ik} + b_5(\text{SR}_{ij} - \text{AR}_i)D + b_6\text{SR}_{ij}, \quad (4)$$

where D is a dummy variable that equals 1 if $\text{SR}_{ij} > \text{AR}_i$, and equals 0 otherwise, and the b's are the coefficients of interest, with $b_6 = 0$.

Before discussing Equation 4, we note that only the USE, COST, and BENEF variables contain k subscripts. Consequently, it is possible to sum over the K different risk-handling activities to obtain the following reduced-form model:

$$\overline{\text{USE}}_{ij} = (b_0 + b_1\text{ML}_i + b_2\text{PRC}_i) - b_3\overline{\text{COST}}_i + b_4\,\overline{\text{BENEF}}_i + b_5(\text{SR}_{ij} - \text{AR}_i)D + b_6\text{SR}_{ij}, \quad (5)$$

where $\overline{\text{USE}}_{ij}$ is the average amount of intended risk-reduction activity, and $\overline{\text{COST}}_i$ and $\overline{\text{BENEF}}_i$ are the associated averages of the perceived costs and benefits for the considered activities. Since Equation 5 has better error properties than Equation 4, while still identifying the coefficients of interest (i.e., b_1 to b_6), we use Equation 5 as the basis to test Hypotheses 1, 2, and 3.

The terms in the first set of parentheses of Equation 5 indicate the extent to which an individual's "normal" shopping activity is affected by individual-level factors (i.e., factors associated with the subject, but not the particular dress). For the special clothing purchases studied here, which manipulate ego involvement and which create purchase involvement because of the nature of the data-collection task, the amount of this search should be significant. Hence, we hypothesize b_0, b_1, and $b_2 > 0$. Hypotheses 1*c* and 1*d* also state that the specific perceived benefit and cost factors should influence the individual's shopping activities. Thus, we postulate $b_3 < 0$ and $b_4 > 0$.

If the level of situation-specific risk perception exceeds the individual's level of acceptable risk, that is, if $(\text{SR} - \text{AR}) > 0$, then the expected amount of risk-reduction activity should increase with this difference. Hence, we expect that $b_5 > 0$ (see Hypothesis 3). If, however, the level of perceived risk for dress j falls below the acceptable level of risk for person i, then the $(\text{SR} - \text{AR})D$ term in Equation 5 drops out (since $D = 0$ when $\text{SR} \leq \text{AR}$). In this case, SR has no effect on use of the various risk-reduction strategies (per Hypothesis 2). To test this hypothesis, we include a separate SR term and postulate that b_6 is zero. (Put differently, we cannot reject Hypotheses 2 and 3 if $b_5 > 0$ and $b_6 = 0$.)

The following section provides a detailed description of both stages of the study. This discussion is organized as follows: (1) subjects and the product chosen for study, (2) operationalizing the theoretical constructs, (3) measurement of the perceived-risk variables (stage 1), and (4) prediction of the normal and extra risk-modification behavior (stage 2).

RESEARCH METHODOLOGY

Subjects and Products Chosen

Subjects participating in the study were a convenience sample of 74 Australian women. Since each respondent completed six separate product evaluations in stage 2, there were potentially 444 observations for estimating Equation 5. Eight of these product evaluations, however, were found to be partially incomplete. Thirty-four percent of respondents were married, 57 percent were in full-time employment, 14 percent were in part-time employment, and 22 percent were university students. The average age of the sample was 30, with ages ranging from 18 to 50.

Dresses were selected because of their association with both uncertain and important consequences. The seasonal impact of fashion change introduces "regular uncertainty" into clothing-purchase decisions. Also, most women perceive clothing-purchase decisions to be important, especially decisions about garments to wear on social occasions. The purchase scenarios used were designed to heighten the uncertainty and importance surrounding purchase of the dresses.

In addition, it was felt that, since women's experience, interest, and self-confidence about buying new outerwear garments varied extensively, both risk perception and the effort devoted to using risk-reducing strategies would also vary widely among women. Finally, because a quasi-experimental methodology was employed, it was important to use a product with which subjects were familiar so as to minimize the artificial nature of the data-collection process.

Operationalizing the Theoretical Constructs

Each construct necessary to test hypotheses 1 to 4 was operationalized by either (1) direct measurement or (2) indirect estimation (e.g., PCR and SR are the by-products of the statistical conjoint model). Factors mentioned in our descriptive model but not measured were controlled for by specifying the amount of the variable to the subject (e.g., buying a dress for a social occasion). Table 1 provides a summary of the operational definitions and the procedure used to quantify each variable. The following sections detail the measures of each variable.

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TABLE 1

ATTRIBUTES USED TO DESCRIBE PRODUCT[a]

Description of attributes	Consequence
Brand	Psychological, social, monetary
Well known to you.	
Not well known to you.	
Self-image	Psychological
Certain that the garment will fit in well with your self-image.	
Uncertain if the garment will fit in with your self-image.	
Style	Functional, psychological, social
Only released for sale in the last couple of weeks.	
Been available since last season.	
Family's and friends' reactions	Social, psychological
Certain your family and friends would approve.	
Uncertain whether your family and friends would approve.	
Certain your family and friends would disapprove.	
Quality	Functional, monetary
Extremely well made.	
Of average quality.	
Not well made.	
Price	Monetary, psychological
$30	
$50	
$90	

[a]The product was a dress.

Measurement of Risk Perception

Measures of risk perceptions typically use a compositional methodology (Dowling 1986). Using this approach, the overall level of perceived risk for a particular multiattribute object (product) is calculated as a weighted sum of the product's perceived attribute levels. Two approaches are generally used to operationalize the components of perceived risk: (*a*) uncertainty multiplied by adverse consequences (based on Bauer's [1960] conceptualization) or (*b*) probability of loss multiplied by importance of loss (popularized by Peter and Tarpey [1975]).

Bettman (1975) noted that it is also possible to use a decompositional methodology to measure perceived risk. This approach decomposes a subject's evaluations of the overall perceived risk of buying a product into the part-worth utilities associated with the attributes of the product. It has the advantages of (*a*) capturing the respondent's overall feeling of risk (which may have both an affective and a cognitive component) and (*b*) providing a method of relating this measure to the specific aspects of the purchase situation. In this study we use a conjoint analysis estimation procedure to develop measures of PCR and SR and thus achieve these advantages.

In stage 1, respondents provided overall perceptions of risk for 24 hypothetical purchase scenarios. To help insure that each respondent had a similar understanding of the concept of risk, the construct was explained to them as the chance that they would suffer adverse consequences as a result of purchasing each dress. Before commencing the rating task, respondents completed two practice profiles that were designed to represent what a priori were believed to be the most and least risky combinations of product attributes. As such, these profiles provided respondents with endpoint anchors for the perceived 15-point risk rating scale, which were labeled "not at all risky" and "extremely risky."

Focus-group interviews were used to guide the selection of the purchase goals and purchase situation (see Exhibit 1), and the six product attributes which were used to describe each hypothetical dress-purchase situation. Each product attribute relates to one or more of the following types of consequences: functional, psychological, social, and monetary, as shown in Table 1. The inclusion of "self-image" and "family's and friends' reactions" as product attributes is an explicit attempt to incorporate the affect component of ego involvement into our risk measures. Also, the use of both positive and negative levels for each attribute (except price) allows perceived risk to be a function of both positive and negative outcomes (a recommendation of Oglethorpe and Monroe [1987]).

Information about each dress was presented to respondents via a full-profile approach. Stimulus profiles were constructed as an orthogonal main-effects, fractional factorial design (Addelman 1962). While this design required only 18 observations, six additional observations were included in the risk-perception task. Three of these extra observations were used to increase the reliability of parameter estimates and three were replicated observations to estimate each respondent's measurement error. Three different random orders of the 24 product descriptions were presented across respondents. Order effects were tested and were generally absent.[1]

Estimation of Product-Category Risk and Specific Risk

Our assumption is that all individuals used a simple additive model of risk perception and that it is composed of two elements, PCR and SR; that is,

[1]The order effect was determined by calculating the importance of each attribute as a function of the particular order of presentation. Importance was measured by calculating for each order the range $U_{max}-U_{min}$, where U is the estimated part-worth of the attribute. These ranges were then compared by testing if the ranges were the same for each order. For each attribute there are three possible comparisons across orders, which yield a total of 18 comparisons. For four attributes, none of the differences in importance were statistically significant. For one attribute, the third order of presentation produced a significantly smaller importance measure than the other two orders. For another attribute, the third order produced a statistically significant lower importance measure than the second order. We take this pattern of results to indicate no significant overall order effect.

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EXHIBIT 1

OPERATIONAL DEFINITIONS FOR VARIABLES IN THE RISK-HANDLING MODEL

Product Attributes	A set of six attributes specified to subjects (see Table 1 for specific attributes and levels; controlled).
Purchase Goals and Purchase Situation	The purchase of a dress to take on a trip to Europe to wear out to a nightclub and other "special occasions." The subject was told she was initially shopping alone, could use a credit card, etc. (controlled).
Potential Consequences	The potential consequences attaching to each product attribute—for example, brand name (not) well known, will (may not) fit in with your self-image, etc. (not directly measured in stage 2, but controlled for by stating attribute levels).
Monetary Costs of Replacing the Garment (ML)	The inability (unwillingness) of the subject to replace the garment if it were inadvertently stolen prior to the trip to Europe (estimated from the responses to three questions).
Overall Perceived Risk (OPR)	The subject's perception of the riskiness associated with buying each dress (measured in stage 1 with a 15-point bipolar scale with the endpoints labeled "not at all risky" and "extremely risky").
Situation-Specific Risk (SR)	The component of OPR that is influenced by the consequences or uncertainty of the situation (estimated by the part-worths in Eq. 6).
Product-Category Risk (PCR)	The subject's overall sensitivity when judging the riskiness of the set of purchase tasks (estimated by the intercept in Eq. 6).
Acceptable Risk (AR)	The amount of product-specific perceived risk above which each subject would like more information before buying the product described to her. (Measured as the breakpoint on 24 perceived risk scales.)
Benefits of Search (BENEF)	The perceived benefit in terms of reducing any risk associated with using a particular risk-modification strategy, such as "shopping with a friend" (measured on a 15-point scale with endpoints labeled "always useful" and "useless").
Cost of Search (COST)	The perceived cost in terms of time, effort, money, or inconvenience associated with using a particular risk-modification strategy, such as "shopping with a friend" (measured on a nine-point scale with endpoints labeled "virtually no cost" and "a large cost").
Search-Strategy Use (USE)	The stated intention to use a specific risk-reduction method (measured on a seven-point scale with endpoints labeled "definitely would [not] use").

$$\mathrm{OPR}_k = \mathrm{PCR} + \sum_{j=i}^{n} \sum_{i=1}^{m_{j-1}} U_{ij} X_{ijk}, \qquad (6)$$

where OPR_k is a (numerical) judgment about the amount of risk perceived to be associated with purchasing the kth dress, with higher values indicating more perceived risk; X_{ijk} is an effects-coded variable representing the ith level of the jth attribute for the kth dress ($X = 1$ if the level is present, -1 if $i = m_j$, 0 otherwise); U_{ij} is a part-worth contribution of the ith level of the jth attribute; n is the number of attributes; m_j is the number of levels within attribute j.

While the form of this perceived risk model is assumed to be the same for all individuals, the part-worths and the constant (i.e., PCR) are assumed to be individually idiosyncratic, reflecting, in part, the influences of the person's wealth, her perceptions of the possible adverse consequences, her desire to avoid risk, and her preference for trading off the disutilities associated with the various product attributes.

The parameter (part-worth) estimation procedure employed is "effects coded" ordinary least squares (OLS) regression (i.e., each attribute is coded as $(m_j - 1)$ independent variables, where the sum of the regression coefficients for a particular attribute is restricted to equal zero, and the intercept term measures a person's average perceived risk level). We take the average risk level across the set of 24 hypothetical purchase scenarios to be our estimate of the person's general PCR.

Measurement of Acceptable Risk

In addition to rating the perceived risk of each product description, respondents stated whether they would be prepared to buy the garment described or whether they would prefer to seek more information prior to making a purchase decision. This response was used in conjunction with the subject's estimated product-specific perceived risk score to calculate the individual's level of acceptable risk. The acceptable risk level was defined as the lowest product-specific perceived risk score associated with a subject's response that she would prefer to seek more information. Alternatively, it can be defined as the largest value of specific risk at which the subject would not require more information. In this way, we assured that acceptable risk was measured on the same scale as product-specific perceived risk.

The calculation of this variable potentially contains two types of error. First, measurement error exists whenever the respondent said she would like more information for a dress that had a lower or equal value of specific perceived risk than a second dress that she was prepared to buy now. Second, experimental (or model-specification) error exists whenever the purchase situations elicited perceived risk levels that did not narrowly bound the acceptable risk value. In the first situation, the average value of all the perceived risk values,

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TABLE 2

SIMPLE EXAMPLE OF HOW STAGE 1 RISK MEASURES ARE ESTIMATED: STIMULUS PROFILE DESIGN AND THE SUBJECT'S RESPONSES

Stimulus profile	Attribute levels			Person 1		Person 2	
	A1	A2	A3	Overall perceived risk rating[a]	Want more info	Overall perceived risk rating[b]	Want more info
1	High	High	High	1	No	3	No
2	Low	Low	Low	13	Yes	13	Yes
3	Low	High	High	3	Yes	7	No
4	High	Low	Low	11	Yes	9	Yes
5	High	Low	High	5	No	7	Yes
6	Low	High	Low	9	Yes	9	Yes

[a]Product-category risk, the average of the obtained responses for the person's overall perceived risk (OPR), is 7.
[b]Product-category risk is 8.

between and including the highest level of risk for which the respondent would buy now and the lowest level of risk for which she would seek more information, is the estimate of acceptable risk. In the second situation, the midpoint of the two risk ratings is used as the estimate of AR.

Since our conceptualization of product-category risk, product-specific perceived risk, and acceptable risk is new, we present a simplified example of how we derived the measures of these constructs. Assume, for simplicity, that two people are given the same six profiles to rate in terms of OPR. In addition, each person is asked to specify those profiles for which she would like to seek additional information prior to making a purchase decision. A hypothetical set of such data is presented in Table 2, which shows an experimental design for such stimulus profiles and the obtained responses for OPR (used to derive estimates for the part-worths) and whether or not the person wanted more information (used to derive estimates of AR). Table 3 gives the estimates of the two respondents' utility values as defined by Equation 6 for each attribute level, and Table 4 indicates the estimates of PCR, perceived SR, and AR.

Note from Tables 2, 3, and 4 that the estimate for PCR is the average of the obtained responses for the person's OPR. Also, estimates of SR for any profile are the sum of the estimated part-worths for that individual for that profile. The AR is measured as the minimum SR value associated with the person's wanting to seek more information. For person 2, this value is clear-cut because the largest "no"-value of SR is the same as the smallest "yes"-value of SR, namely, -1. However, for person 1, the acceptable risk value is measured with some error and is the midpoint between the largest no-value (-2) and the smallest yes-value (-4).

Costs, Benefits, and Monetary Loss

Respondents in stage 1 also rated the perceived cost and benefits associated with each of the 25 risk-modification strategies used in stage 2. (These are described in the next section.) Costs were measured on a nine-point scale anchored by the phrases "virtually no cost" and "a large cost." Benefits were measured on a 15-point scale anchored by "always useful" and "useless." The variables $\overline{\text{COST}}$ and $\overline{\text{BENEF}}$ were obtained by summing the appropriate responses for the respective questions and averaging.

Finally, each respondent answered three questions aimed at measuring her inability to incur a monetary loss associated with the purchase of a dress similar to the ones she evaluated in the perceived risk-measurement scenarios of stage 1. Respondents were asked what they would do if, after purchasing the $30 dress for a trip to Europe, the dress was stolen. Permissible actions were (1) buy another identical garment, (2) replace it with a cheaper version, or (3) not replace the garment. The same question also was asked for dresses costing $50 and $90. Responses to these three ordered categories indicate the respondent's ability to absorb the monetary loss associated with each dress. We used the simple transformation of assigning each category an evenly spaced numerical value that preserved the original order (viz., 1, 2, and 3) and summed these values to produce our measure of each respondent's inability to suffer an ML associated with buying the type of dress used in this study (Dowling and Midgley 1991). The scale had an acceptable Cronbach α of .75 (Nunnally 1978). Higher values of ML indicate a greater inability to suffer such losses.

Evaluating the New Product Stimuli and Measuring Risk-Modification Behavior

Immediately after completing stage 1, respondents were asked to examine three groups of 30 color magazine photographs of women's dresses. Each group represented a different priced dress; the price categories used were $30, $50, and $90, the same as those in the first stage of the study. After being informed of the price

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TABLE 3

SIMPLE EXAMPLES OF HOW STAGE 1 RISK MEASURES ARE ESTIMATED: RESPONDENTS' DERIVED ATTRIBUTE UTILITY VALUES

	Attribute Level	Person 1's utility values	Person 2's utility values
A1	High	−1	−2
	Low	1	2
A2	High	−2	−2
	Low	2	2
A3	High	−3	−1
	Low	3	1

TABLE 4

SIMPLE EXAMPLES OF HOW STAGE 1 RISK MEASURES ARE ESTIMATED: RISK MEASURES

Stimulus profile	Person 1's product-specific risk[a]	Person 2's product-specific risk[b]
1	−6	−5
2	6	5
3	−4	−1
4	4	1
5	−2	−1
6	2	1

[a]Acceptable risk is calculated to be −3.
[b]Acceptable risk is calculated to be −1.

associated with each category, respondents were instructed to choose two dresses from each group, one dress that they were certain would fit in well with their self-image, and a second that they were not certain was compatible with their self-image. These levels are identical to the levels used to measure self-image in stage 1 (see Table 1).

The selection of six photographs by respondents served a twofold purpose. First, the photograph-selection task acted as an enjoyable interlude between the completion of two sets of paper and pencil questionnaires and, thus, tended to disassociate the stage 1 risk-assessment measurements from the stage 2 risk-handling measurements. Second, the six photographs selected provided a realistic way of presenting the product stimuli to the respondents, thereby reducing possible demand effects.

After selecting the photographs, respondents rated each of the six dresses on the two other perception variables used in the first stage, namely, the length of time they thought the style had been on the market and the likely reaction of their family and friends if they purchased the dress.[2] Use of this information defines specific levels for four of the six attributes that were postulated to cause perceived risk (i.e., price, self-image, style, family's and friends' reactions). The pictures, price and self-image variables defined six different purchase situations. For each, respondents completed 21 questions describing their likely information search and shopping behavior, with the assumption that they were interested in buying a dress similar to the one in the picture that they had chosen. The extent of intended use of each search strategy (USE_{ijk}) was measured by a seven-point scale with endpoints labeled "definitely (would/not) use."

[2]Respondents rated the style variable on a six-point scale with endpoints the same as those in Exhibit 1. A response in the range 1–3 was classified (by the researchers) as "available last season" while a rating of 4–6 was classified as "only released for sale in the last couple of weeks." The family and friends reaction variable was rated on a seven-point scale, where the values 1–2 were classified as "certainly approve of your choice," 3–5 as "uncertain," and 6–7 as "certainly disapprove of your choice."

The search-strategy questions were arranged as a structured protocol and are shown in the left-hand portion of Table 5. For example, the first eight questions dealt with activities a respondent could undertake before going out to look for the dress, the next four questions dealt with the search for a retailer, and the next nine questions asked what the respondent would do when she found a shop with a dress like the one in the photograph. These 21 questions covered the areas of retailer, media, interpersonal, and introspective search, as detailed by the right-hand portion of Table 5. Use of any particular strategy could reduce the uncertainty and/or possible adverse consequences associated with one or more of the types of loss noted in Table 1.

After completing these 21 questions, respondents were told that the dress was of one of the three different quality levels in Table 1 and that the brand name was either well known or not known to them. (These two pieces of information would normally be gained during the shopping process covered by the first 21 questions.) Four more questions concerning likely shopping behavior were then completed (see questions 22–25 in Table 5). Each respondent answered a total of 150 (25 × 6) search strategy questions. At the completion of the total data-collection task, each respondent was paid a small monetary fee, asked about her feelings while completing the task, and debriefed. Most women said that they enjoyed looking at the photos, which we consider to be a natural outcome of their purchase involvement. Also, no respondent mentioned that she thought that her answers in the first questionnaire influenced those in the second questionnaire.

Since it is possible that respondents took different times to "learn" about the risk-handling procedures available while evaluating the six product scenarios, four different orders of presentation of the six dress stimuli were used. As a result, each product scenario had the same average order of presentation to respondents. The learning hypothesis was tested by using a two-factor ANOVA, where order of presentation and the product scenario were the independent variables and the sum

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TABLE 5

RISK-REDUCTION STRATEGIES AND INFORMATION SOURCES

Strategy	Information source			
	Retailer	Media	Interpersonal	Introspection
Before going out to look for a dress, would you:				
1. spend time thinking about whether the dress will suit you?				✓
2. ask your family and friends for their advice?			✓	
3. read the fashion section of magazines and newspapers for information about whether the style is in fashion this season?		✓		
Would you look at advertisements to see:				
4. which shops are selling similar dresses?		✓		
5. whether any shop has a dress like this one on sale?		✓		
6. spend time thinking about what color, material, etc. to get the dress in?				✓
7. wait until several people you know have bought a similar style?			✓	
8. wait until you have seen someone else wearing the garment before you buy it?			✓	
While out shopping for a dress, would you:				
9. take time to look at clothing displays in shop windows?	✓			
10. go to the one or two shops you are most familiar with to look at their range of dresses?	✓			
11. also go to a number of unfamiliar shops to look at their range of dresses?	✓			
12. shop for a dress with a family member or friend?			✓	
Having found a shop with a dress you liked, would you talk to salespeople for their advice about:				
13. whether they think the style suits you?	✓			
14. how to care for the dress?	✓			
15. whether the dress is good value for the money?	✓			
16. whether the dress is in fashion?	✓			
17. how many of these dresses have been sold?	✓			
Would you:				
18. use the price of the dress as a guide to its quality?				✓
19. try on the dress to see if it suits you?				✓
20. examine the dress to see if there are any loose threads, the zipper and/or buttons are attached securely, there are no mistakes in the seams, etc?				✓
21. put the dress in layaway?	✓			
After the salesperson told you that you could not return or exchange the dress, would you:				
22. look for another dress in this shop that was made by a well-known manufacturer?	✓			
23. ask if the dress has a manufacturer's guarantee against faulty workmanship?	✓			
24. look for a similar dress in another shop where you could exchange it, if after buying the dress you decided you didn't like it?	✓			
25. look for a similar dress in another shop where you could return it for cash?	✓			

of the 25 risk-handling scores was the dependent variable. An order effect was found ($F = 3.26$, $p = .01$), but there was no interaction with the scenario factor ($F = 0.32$, $p = .99$). While the order effect was not totally unexpected, it posed no problem to testing the hypotheses since the lack of any interaction between order of presentation and scenario indicates that the order effects are the same for each product stimulus and may effectively be ignored (Keppel 1973, pp. 395–400).

Estimating the Perceived Risk of the New Product Stimuli

To test Hypotheses 1, 2, and 3, the part-worth disutilities (estimated from stage 1) corresponding to the attribute levels used to describe the six dresses were used to estimate the specific risk perceived. Two such estimates of this perceived risk were calculated for each dress (for each respondent). The first estimate was based on the four pieces of information available to respondents at the beginning of the "shopping trip," namely, price, self-image, family's and friends' reactions, and style. For this first estimate of specific perceived risk it was assumed that the values of the part-worths for the two remaining bits of information were zero, because respondents were unlikely to form a strong opinion about the quality of the dress or its brand name from the information presented to them.[3] This first value of

[3]We believe this to be true for three reasons. First, in the absence of any information the expected part-worth for any attribute is zero under the effects-coding scheme used here. Second, while subjects may make some inference about quality from the price information given, price is not a strong signal of product quality in this market (Griffin and O'Neal 1992). Third, it is unlikely that subjects would make inferences about how well known the brand name is from the first four bits of information.

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product-specific perceived risk (SR1) was used to predict the average of respondents' intended use of the first 21 risk-reduction strategies ($\overline{USE1}$). By summing and averaging across the 21 risk-reduction strategies, this variable reflects both the number and intensity of risk-modification strategies and is similar to measures used by Brown and Gentry (1975), Herman and Locander (1977), Lutz and Reilly (1973), Murray (1991), and Rao and Farley (1987).

A second value of each situation's product-specific perceived risk (SR2) was obtained by including the relevant part-worth values of the new pieces of information (i.e., quality and brand name) provided in the latter part of the shopping task. This second value was used to predict the average of respondents' intended use of the final four risk-reduction strategies ($\overline{USE2}$). Having two sets of specific perceived risk and risk reduction values allows us to test Equation 5 twice and to (partially) determine the robustness of the variable relationships in Hypotheses 1, 2, and 3 to the consumer acquiring new information.

RESULTS

Reliability and Validity Checks of Risk Variables

Before testing the hypotheses we checked the reliability and validity of the product-category risk sensitivity and specific-risk measures derived from respondents' overall perceived risk evaluations in stage 1. If a significant proportion of respondents did not map the factors in the top portion of Figure 1 into perceived risk, then we would be unable to test the hypotheses. The formal testing procedure used the six product features describing the 24 hypothetical dress-purchase scenarios as effects-coded variables in an OLS regression for each respondent. If these six factors and the intercept term of the regression equation explained a significant proportion of the variance in a respondent's overall perceived risk score, then we could proceed. Other procedures then could be employed to lend further support to our risk measures.

Two decision rules were used, against which each respondent's overall risk ratings were checked. Failure to pass either decision rule resulted in the respondent's violating minimal reliability and validity conditions. These respondents were called the "poor model-fit group." The first decision rule used was the degree of fit of the model estimated by Equation 6. Any respondent whose disutility model (i.e., individual OLS regression) was not significant at $p = .10$ was assigned to the poor model-fit group. Seven of the 74 respondents violated this criterion.

The second decision rule, which in essence is a reliability criterion, was based on the three replicated hypothetical product scenarios evaluated by each respondent. Any respondent whose average standard deviation across these replications was greater than 3 SD above the average of all respondents was assigned to the poor model-fit group. One new respondent violated this criterion.[4] Eight respondents in total (or 11 percent) were classified in the poor model-fit group.[5] Because these respondents violated the assumptions inherent in the measures of perceived risk used to test the hypotheses, it was decided that any attempt to predict risk-handling behavior from this data would be meaningless. To support this decision, a test of the equality of the risk-induced shopping behavior of both groups of respondents was conducted. The results (reported later in this section) showed significant differences between the response profiles of both groups.

Two internal validity checks were performed on our risk measures. First, we calculated the coefficient of multiple determination of each individual regression; R^2s ranged from .60 to .99 and averaged .83 across all respondents. A second internal validity check was to evaluate the "face validity" of the part-worths from these individual regressions (Scott and Wright 1976). Only 3.6 percent of the total number of parameters estimated across all respondents had signs and/or magnitudes contrary to prior expectations. Finally, we calculated the number of attributes used by each respondent. Attribute salience was estimated with a main-effects fractional factorial ANOVA. Any main effect significant at $p = .10$ was defined as a salient attribute. Respondents used between two and six of these in the risk-perception task, with an average of 3.1. This evidence suggests that respondents behaved in a manner compatible with the premise that they traded off among attribute values to determine perceived risk and that they had an adequate number of cues to evaluate this risk.

In sum, these results support the premise that a person's perception of the riskiness of a purchase situation can be decomposed into an individual-specific risk factor and a factor associated with the uncertainties and potential adverse consequences of the product-category situation. Whether our inability to model risk perception for the 11 percent of our sample who fell into the poor model-fit group was due to its members' inability (or unwillingness) to evaluate the riskiness of the stimuli, or because risk perception for this group of respon-

[4]The average test-retest reliability of the OPR measures for the 66 respondents on which Hypotheses 1–4 were tested was .65 and ranged from .48 to .82. Most of this unreliability is attributed to the quite difficult (and unfamiliar) task of integrating up to six pieces of information to form an overall perceived risk judgment.

[5]An interesting feature of this group's responses is that only an average of 1.1 product attributes were found to be significant in the perceived-risk evaluation task. It seems reasonable to assume that this group either used information not provided to evaluate the perceived riskiness of the set of hypothetical scenarios or were unable (or unwilling) to expend the cognitive effort to evaluate each scenario in terms of its riskiness. For the purpose of this study their responses are unreliable.

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TABLE 6

REGRESSION COEFFICIENTS FOR THE MODEL OF RISK REDUCTION

	$\overline{USE1}$				$\overline{USE2}$			
	β	VIF	t	Probability[a]	β	VIF	t	Probability[b]
ML	.22	1.2	4.95	.00	.21	1.2	4.48	.00
PCR	.02	1.5	.40	.69	.02	1.3	.47	.71
$\overline{COST}$	−.00	1.2	−.02	.99	.02	1.1	.41	.68
$\overline{BENEF}$	.29	1.2	6.62	.00	.40	1.1	8.62	.00
(SR − AR)D	.38	4.3	4.49	.00	.24	3.4	3.03	.00
SR	.00	3.7	.06	.96	.09	3.3	.79	.27

NOTE.—The term $\overline{USE1}$ is the average of the intended use of the first 21 risk-reduction strategies (Table 5); $\overline{USE2}$ is the average of the intended use of the final four risk-reduction strategies; ML is monetary loss; PCR is product-category risk; $\overline{COST}$ is the associated average of the perceived costs; $\overline{BENEF}$ is the associated average of the perceived benefits; SR is product-specific risk; AR is acceptable risk; D is a dummy variables that equals 1 if $SR_{ij} > AR_i$ and that equals 0 in all other cases; VIF is the variance inflation factor.

[a]Adjusted R^2 is .36; $F(6, 381)$ is 37.8.

[b]Adjusted R^2 is .36; $F(6, 316)$ is 31.7.

dents is not associated with the situation specific attributes, is left to future research.

When compared to much of the empirical research on risk perception, the analyses discussed above provide new insight into the claim that people can think in terms of risk, and that this perceived risk is influenced by characteristics of the purchase situation. Most prior work in this area uses (compositional) self-completed measures of risk perception and frames questions to subjects explicitly as potential losses. These evaluations are then *assumed* to measure perceived risk. Our determinants of risk are framed in more neutral terms (Table 1), making the link between the product attributes and perceived risk less obvious. Hence, we take the above results as substantive evidence that most people can readily and consistently think in terms of perceived risk and that this risk is composed of a fixed component (product-category risk) and a variable component (product-specific risk).

Test-Retest Reliability of the Risk-handling Measure

To estimate the measurement error associated with the six sets of the 25 risk-handling questions evaluated by each respondent, a subsample of 10 respondents completed the second stage of the study again two weeks later. Variation in risk-handling scores over time can be attributed to random factors, such as respondent mood when completing the task and/or inherent changes in shopping (search) patterns. While these two causes of variance cannot be separated, changes in mood rather than genuine changes in shopping behavior should be the dominant cause. Correlations between the summation of the two sets of risk-handling scores collected for the two time periods were .82 (for $\overline{USE1}$) and .79 (for $\overline{USE2}$), both significant at $p = .01$. We take these results to indicate that our measures were acceptably reliable; that is, respondents had a consistent schema of what they would do when faced with a particular situation.

Relationship among the Individual-specific Risk Factors

Hypothesis 4 predicts that acceptable risk is negatively related to product-category risk and the inability to suffer a monetary loss. That is, these three individual-specific risk factors all reflect the individual's general sensitivity to risk. To test the significance of this relationship, we regress acceptable risk against product-category risk and monetary loss. The F-value for this regression for the 66 good respondents is 52.2, which is significant at $p = .00$. The β coefficients for the two independent variables are highly significant ($p = .00$) with signs in the predicted direction. We take these results as supportive of Hypothesis 4.

Testing the Risk-handling Model

Hypothesis 1 implies significant (and positive) coefficients for PCR, ML, and $\overline{BENEF}$ and a significant (and negative) coefficient for $\overline{COST}$ in Equations 4 and 5. Hypothesis 2 implies that the coefficient for the separate product-specific risk term (i.e., b_6) is not significantly different from zero, while Hypothesis 3 implies a positive and significant coefficient on the term representing the amount of product-specific risk that exceeds the person's acceptable risk (i.e., b_5). This latter hypothesis is a key aspect of our model.

Ordinary least squares regression was used to estimate Equation 5 for each of the two dependent variables, $\overline{USE1}$ and $\overline{USE2}$. Table 6 shows the regression results for Equation 5 and indicates a significant statistical relationship between both risk-handling variables and the

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set of independent variables.[6] In all cases, the signs of the regression coefficients are in the predicted direction except for the statistically insignificant COST variable in the USE2 equation. Before interpreting the regression coefficients, however, we followed the suggestions of Mason and Perreault (1991) and checked for small sample size, low overall model fit, or extreme intercorrelation of predictor variables, any one of which can adversely affect the accuracy of regression coefficients. Because we used their standards, our results are based on a big sample size and have low to moderate overall model fit (as determined by R^2). We assess the degree of collinearity among our predictor variables by computing a variance inflation factor (VIF) for each predictor variable.[7] Marquardt (1970) suggests that any VIF greater than 10 signals harmful collinearity. Table 6 shows that the highest VIF equals 4.3. On the basis of the size of our sample, R^2 and the highest VIF, our conclusion is one of general confidence in inference.

Table 6 shows a high degree of correspondence between the results for the two measures of risk handling. In both cases respondents who indicated their inability to absorb a monetary loss (significant ML coefficient) also reported more intended shopping behavior. This finding lends support to Hypothesis 1*b*. In addition, although the coefficients on COST are insignificant, the coefficients for BENEF are positive and highly significant, which indicates that the respondent's perception of the benefits of a particular risk-reduction strategy will influence its intended use.

Also of interest is the lack of significance of product-category risk in both equations. While this measure is correlated with monetary loss ($r = .27$, $p < .01$), each construct seems to be tapping a different aspect of a person's general risk-attitude profile. These results seem to indicate that, after adjusting for ability to absorb a monetary loss, a person's purchase risk sensitivity does not directly affect intended shopping activity. We are unable to ascertain whether the less economically based measure has no influence or whether its influence is through the setting of acceptable risk.

The tests for Hypotheses 2 and 3 revolve around the coefficients for SR and (SR − AR)*D*. As can be seen from Table 6, the significant coefficients for (SR − AR)*D* and the insignificant coefficients for SR for both dependent measures give credibility to Hypotheses 2 and 3. Hence, there seems to exist an acceptable level of risk below which the effects of specific perceived risk are minimal and above which increases in specific perceived risk have a substantial effect on the person's intended risk-reduction behavior.

[6]Owing to a typing mistake on the questionnaires, the responses to one of the six dresses for each respondent had to be eliminated for the analysis of USE2. Also, because of missing data on some sets of risk-handling questions, the degrees of freedom vary between USE1 and USE2.

[7]The VIF = $(1 - R_k^2)^{-1}$, where R_k^2 is computed by regressing each predictor variable, one at a time, on the other predictor variables.

Finally, to support our decision to drop the poor model-fit group of respondents, we conducted a test of the equality of the regression lines (intercepts and slopes) between groups (Neter and Wasserman 1974, pp. 160–165). For both dependent variables the groups were significantly different: USE1, $F = 6.55$, $df = 6,422$, $p = .00$; USE2, $F = 2.04$, $df = 6,358$, $p = 0.00$. Also, the signs of some of the slope coefficients for the poor model-fit group were not in the hypothesized direction. These results, in concert with those reported earlier, support our decision to exclude this group from the hypothesis-testing stage of the study.

DISCUSSION

The accurate measurement of perceived risk and risk-reduction activities is a difficult task. We propose that risk perception contains a cognitive and an affective component, and our measures of risk (i.e., product-category risk, product-specific risk, acceptable risk and ability to suffer a monetary loss) sought to capture both these aspects. Also, our experimental design ensured that the measurement of these constructs was independent of the measurement of the risk-reduction activity they were designed to predict. This was intended to provide a tough test of our model.

The measures of risk reduction were obtained by asking respondents to predict the type of behavior that they would engage in rather than monitoring their actual shopping behavior. From one perspective, not measuring actual behavior could be viewed as a deficiency in our research design. However, in this study we were less interested in what people actually do than in what they plan to do, since the former will be greatly influenced by the specific information obtained as the search process progresses. By focusing on intended behavior, we more faithfully represent the effects of risk perception on expected risk-reduction behavior.

The major finding of our study is that product-specific perceived risk can greatly affect the intensity of the intended use of risk-modification strategies and that there appears to be an acceptance value below which this product-specific risk has little effect on intended behavior. We also find that respondents' use of various shopping activities appears to be positively influenced by their perceived benefit of the activity to reduce perceived risk. It is interesting that the perceived costs of using these activities appear to have little influence on intended use. Perhaps our measure of cost of use was poor, or our design did not impose a cost for respondents that could affect their intended use to any significant extent. An alternative explanation is that the cost of using any particular activity might have been so low compared to the perceived benefit that the cost component was ignored. Stated differently, when the perceived benefits are high relative to the perceived costs, it may be that the benefits completely dominate the costs. This could easily happen in our situation, since

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we were dealing with situations perceived to be relatively high in risk and the costs of using any activity were quite low.

The implications of this study's findings can be assessed from both a theoretical and a practical perspective. First, our model of risk perception indicates that most respondents had little trouble mapping the potential positive and negative outcomes of a purchase situation into ratings of perceived risk. As we noted earlier, the compositional measures of perceived risk most popular among consumer researchers seldom explicitly test this assumption. An interesting research question is whether perceived risk measured using a decompositional approach, such as the one used here, would correlate highly with a compositional measure.

A second theoretical finding is that, like Bettman (1973, 1975), we also find that, when evaluating products in a product class, respondents' risk-reduction behavior is primarily motivated by the specific perceived risk associated with the particular product being considered (if it is above their acceptance level). Thus, even though the risk associated with the product class (product-category risk or Bettman's inherent risk) may influence basic patterns of risk-reduction behavior across different types of product classes, it seems to have little effect when a person is searching for a particular product in a product class. To our knowledge, this is one of the first consumer-research studies to empirically assess the effect of an acceptance level of risk perception on any type of consumer behavior. While our measure of this construct contained some error, the findings reported in this article suggest that incorporating such threshold-type variables into models of risk perception and handling is worthwhile. In this study, the acceptable-risk variable helped us to better understand the effects of perceived risk on information search and other forms of shopping activity. Future research could focus on replicating our findings and calibrating the effects of such a variable on consumers' decisions to exclude a product from their choice set.

The final theoretical insight is the role of an individual's wealth position as a motivating force for his/her product-purchase behavior. Psychologists and economists have argued for many years that a person's current and anticipated wealth position will influence the way they value alternatives (see, e.g., Kahneman and Tversky 1979). Here we use a rather specific measure of wealth, namely, the inability to absorb a monetary loss associated with a purchase, and find that it is a powerful determinant of risk-reduction activity. While many perceived-risk constructs incorporate implicit wealth effects in their measures (e.g., the importance of the loss component of compositional measures), few researchers attempt to explicitly model this aspect of a person's disposition. Future research could focus on developing a more general wealth construct that incorporates both monetary and nonmonetary aspects.

Finally, we leave to future research a test of our premise that consumers continue to seek information until they are able to reduce their specific risk to a level below their acceptable risk. Such an investigation will require a longitudinal design versus the cross-situation design employed herein.

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